How one reconstruction is produced

Follow the numbered steps. Feedback paths show what each optimizer changes.

Observed measurement
y · 64×64

1 Inputs
ε: continuous noise
s: 32 discrete styles
c: target class

2 Generator
Gθ(ε, s; c)
pretrained 134M weights
one forward pass

3 Candidate
proposed HR image
x = Gθ(ε, s; c)
256×256 pixels

4 Sensor model
predicted Ax · 64×64
bicubic 4× downsample
compare with observed y

5 Loss
‖y − Ax‖²
drives optimization
no HR target is used

IADrift: also update θ

CSGM: update ε only

What each method adds

Search, combine, then enforce measurement consistency.

1 Search and adapt

CSGM
Optimize ε · 5,000 steps · 2 restarts; keep lower LR loss.

Style-label search
Try 16 labels for 500 steps; finish best 2; keep better result.

IADrift
Continue from CSGM; update θ + ε; read outputs at 100 and 1,000 steps.

2 Combine solutions

Ens-U / Ens-W
Average 5 CSGM solutions: uniform or weighted by LR loss.

Label ensemble (dev)
Average up to 12 starts and labels; optionally continue each with IADrift@100.

3 Enforce

Back-projection
force Ax = y
closest consistent image for every method

Notation: code uses z (‘noise_labels’) for s. Full hyperparameters remain in the paper text and caption.